TV Intelligence Scanner — Firmware Upgrade Report

Session #1 | TV project #1 SAMPLE — Bench TV (synthetic data) — Acme TESTTV55 | method official_usb (profile #1, verified: yes, root) | target firmware #2 upgrade_tv_v2.bin SHA-256 0706d080792e18981c0f257142fec834d5391eb9ccd0c10a471c3b8b1e017ebe | from version V1.0 | status recovery:done

Documentation: Acme TESTTV user manual (synthetic) p.40 'Software update via USB'

Steps

Step	Status	Details
identification	warning	{"values": {"model": "TESTTV55", "mainboard": "TP.MS338.PB801", "chassis": "CH-9", "pcb_revision": null, "soc": "MSD6A648", "panel": "LC550EGE", "current_firmware": "V1.0", "identification_status": "identified", "target_firmware": "upgrade_tv_v2.bin"}, "missing_required": [], "missing_optional": ["pcb_revision"]}
compatibility	passed	{"verdict": "VERIFIED COMPATIBLE", "score": 70, "reasons": ["model, architecture, board/SoC and layout confirmed against the TV's own firmware; no concerns"], "compatibility_id": 2}
validation	passed	{"checks": [{"name": "SHA-256", "ok": true, "detail": "0706d080792e18981c0f257142fec834d5391eb9ccd0c10a471c3b8b1e017ebe"}, {"name": "File integrity", "ok": true, "detail": "verified"}, {"name": "Package name vs documentation", "ok": true, "detail": "upgrade_tv_v2.bin vs upgrade_*.bin"}, {"name": "Firmware metadata", "ok": true, "detail": "analysis run #2, 19 facts"}, {"name": "File format", "ok": true, "detail": "Unidentified binary (.bin)"}, {"name": "Digital signature", "ok": null, "detail": "Android Verified Boot, SHA256_RSA2048 (presence only; keys not verified)"}, {"name": "Source reliability", "ok": true, "detail": "Official; downloaded; SHA-256 matches the value published by the source"}, {"name": "Compatibility report", "ok": true, "detail": "VERIFIED COMPATIBLE (report #2)"}]}
backup	done	{"items": {"original_firmware_version": "V1.0", "full_firmware_image": "YES \u2014 library firmware #1 orig_dump.bin (SHA-256 cce88a2118fc3fd1\u2026)", "configuration_export": "Settings exported via official menu (synthetic)", "hardware_report": "/tmp/tmp2qb6piv8/data/storage/reports/1/Hardware_Report_SAMPLE \u2014 Bench TV (synthetic data)_20261108_220706_390276.pdf", "notes": ""}}
preparation	passed	{"checks": [{"name": "Stable power (UPS / no storm)", "ok": true, "detail": "Do not switch off during the update"}, {"name": "Connection / medium ready", "ok": true, "detail": ""}, {"name": "Filesystem", "ok": true, "detail": "FAT32 (documented: FAT32)"}, {"name": "Free space", "ok": true, "detail": "10,000,000,000 B free, 1,848,179 B needed"}, {"name": "Removable drive", "ok": true, "detail": "removable"}, {"name": "Package file name", "ok": true, "detail": "upgrade_tv_v2.bin vs documented pattern upgrade_*.bin"}, {"name": "Target path", "ok": true, "detail": "/tmp/tmp2qb6piv8/work/USB/upgrade_tv_v2.bin"}, {"name": "User confirmation", "ok": true, "detail": "confirmed by root"}], "risks": ["Method 'Official USB Upgrade' \u2014 risk level medium. Prepares a USB stick exactly as documented; the TV's own updater installs it.", "Power loss or removing the medium during the TV's update can make it unbootable.", "Rollback: NOT documented for this model; do not assume the previous firmware can be restored.", "Compatibility: VERIFIED COMPATIBLE", " \u2022 model, architecture, board/SoC and layout confirmed against the TV's own firmware; no concerns"], "method": "Official USB Upgrade", "usb_target": "/tmp/tmp2qb6piv8/work/USB", "instructions": ["Method: Official USB Upgrade. Follow the manufacturer's documentation: Acme TESTTV user manual (synthetic) p.40 'Software update via USB'.", "Power: Do not switch off during the update", "Do NOT switch off the TV or remove the medium while t
execution	done	{"executed": true, "ok": true, "usb_file": "/tmp/tmp2qb6piv8/work/USB/upgrade_tv_v2.bin", "sha256": "0706d080792e18981c0f257142fec834d5391eb9ccd0c10a471c3b8b1e017ebe", "bytes": 726912, "instructions": ["Method: Official USB Upgrade. Follow the manufacturer's documentation: Acme TESTTV user manual (synthetic) p.40 'Software update via USB'.", "Power: Do not switch off during the update", "Do NOT switch off the TV or remove the medium while the TV is updating."], "next": "Insert the USB stick into the TV and follow the documented steps. Do not switch off the TV while it updates. Then record the result in the verification step."}
verification	passed	{"reported_version": "V2.0", "expected_version": "V2.0", "version_check": "matches", "model_ok": true, "panel_ok": true, "boot_ok": true, "functions": {"picture": true, "sound": true, "remote": true, "network": true}, "failed_functions": [], "notes": ""}
recovery	done	{"guidance": ["Documented recovery (Acme TESTTV user manual (synthetic) p.40 'Software update via USB'):", "If the TV does not start, contact Acme service (synthetic)", "Rollback: NOT documented \u2014 the previous firmware may not be installable.", "Saved before upgrade: version V1.0; full image: YES \u2014 library firmware #1 orig_dump.bin (SHA-256 cce88a2118fc3fd1\u2026)"]}

Log

[2026-11-08T22:07:06Z] Session started: method official_usb, profile #1, target firmware #2 [2026-11-08T22:07:06Z] Step identification: warning [2026-11-08T22:07:06Z] Step compatibility: passed [2026-11-08T22:07:06Z] Step validation: passed [2026-11-08T22:07:06Z] Step backup: done [2026-11-08T22:07:06Z] Step preparation: passed [2026-11-08T22:07:06Z] Copying upgrade_tv_v2.bin (726,912 B) to /tmp/tmp2qb6piv8/work/USB/upgrade_tv_v2.bin [2026-11-08T22:07:06Z] Verifying copy (SHA-256)… [2026-11-08T22:07:06Z] SHA-256 on USB: 0706d080792e18981c0f257142fec834d5391eb9ccd0c10a471c3b8b1e017ebe — MATCH [2026-11-08T22:07:06Z] Step execution: done [2026-11-08T22:07:06Z] Step verification: passed [2026-11-08T22:07:06Z] Step recovery: done

This application never writes to the TV, its flash memory, partitions or bootloader. With the Official USB method it only prepares the USB medium exactly as documented and verifies the copy; the TV's own updater performs the installation.